

pantoprazole (Rx)

Brand and Other Names: Protonix

Classes: Proton Pump Inhibitors



Dosing & Uses

ADULT

Dosage Forms & Strengths

tablet, delayed-release

- 20mg
- 40mg

granules for oral suspension, delayed-release

- 40mg/packet

injection, solution (single-dose container)

- 40mg/100mL in 0.9% NaCl (0.4mg/mL)
- 40mg/50mL in 0.9% NaCl (0.8mg/mL)
- 80mg/100mL in 0.9% NaCl (0.8mg/mL)

injection, powder for reconstitution

- 40mg/vial

Erosive Esophagitis Associated With GERD

Treatment: 40 mg PO qDay for 8-16 weeks

Maintenance of healing: 40 mg PO qDay

Alternatively, 40 mg IV qDay for 7-10 days

Short-term Treatment of GERD

Oral therapy inappropriate or not possible: 40 mg IV infusion over 15 minutes qDay for 7-10 days; switch to PO once patient able to swallow

Zollinger-Ellison Syndrome

40 mg PO qDay; up to 240 mg/day administered in some patients

80 mg IV infusion q8-12hr up to 7 days; switch to PO once patient able to swallow

Peptic Ulcer Disease (Off-label)

Duodenal ulcer: 40 mg PO qDay for 2 weeks

Gastric ulcer: 40 mg PO qDay for 4 weeks

PEDIATRIC

Dosage Forms & Strengths

granules for oral suspension, delayed-release

- 40mg/packet

tablet, delayed-release

- 20mg
- 40mg

injection, powder for reconstitution

- 40mg/vial

Erosive Esophagitis Associated With GERD

Oral

- Indicated for short-term treatment (up to 8 weeks) of erosive esophagitis (EE) associated with gastroesophageal reflux disease (GERD) in patients aged ≥ 5 years
- < 5 years: Safety and efficacy not established
- ≥ 5 years
 - 15 kg to < 40 kg: 20 mg PO qDay
 - ≥ 40 kg: 40 mg PO qDay

- Not to exceed 8 weeks duration

IV

- Indicated for short-term treatment (up to 7 days) of EE associated with GERD in children aged ≥ 3 months
- < 3 months: Safety and efficacy not established
- Discontinue IV administration as soon as patient is able to tolerate oral treatment
- 3 months to < 1 year
 - < 12.5 kg: 0.8 mg/kg IV qDay
 - ≥ 12.5 kg: 10 mg IV qDay
- 1-17 years
 - ≤ 15 kg: 10 mg IV qDay
 - > 15 kg to ≤ 40 kg: 20 mg IV qDay
 - > 40 kg: 40 mg IV qDay

Interactions

Interaction Checker

Enter a drug name to check for any interactions. and

pantoprazole

No Interactions Found

Interactions Found

Contraindicated

Serious

Significant - Monitor Closely

Minor

All Interactions Sort By:

Contraindicated (2)

- nelfinavir
pantoprazole decreases levels of nelfinavir by unspecified interaction mechanism. Contraindicated. Coadministration may lead to loss of nelfinavir virologic response and development of resistance; mechanism may be CYP2C19 inhibition of nelfinavir conversion to active M8 metabolite, and also PPIs decreasing gastric pH resulting in decreased nelfinavir absorption.
- rilpivirine

pantoprazole decreases levels of rilpivirine by increasing gastric pH. Applies only to oral form of both agents. Contraindicated. Concurrent use may cause treatment failure and/or the development of rilpivirine or NNRTI resistance owing to decreased levels.

Serious (32)

- **acalabrutinib**
pantoprazole decreases levels of acalabrutinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Acalabrutinib solubility decreases with increasing gastric pH. Due to the long-lasting effect of PPIs, separation of doses may not eliminate the interaction.
- **alpelisib**
pantoprazole will increase the level or effect of alpelisib by Other (see comment). Avoid or Use Alternate Drug. Coadministration of alpelisib (BCRP substrate) with a BCRP inhibitor may increase alpelisib concentration, which may increase the risk of toxicities. If unable to avoid or use alternant drugs, closely monitor for increased adverse reactions.
- **apalutamide**
apalutamide will decrease the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Avoid or Use Alternate Drug. Coadministration of apalutamide, a strong CYP2C19 inducer, with drugs that are CYP2C19 substrates can result in lower exposure to these medications. Avoid or substitute another drug for these medications when possible. Evaluate for loss of therapeutic effect if medication must be coadministered.
- **atazanavir**
pantoprazole will decrease the level or effect of atazanavir by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Atazanavir solubility decreases as pH increases. Substantially reduced plasma concentrations of atazanavir are expected if PPIs are coadministered. PPI dose should not exceed a dose comparable to omeprazole 20 mg and must be taken ~12 h before atazanavir/ritonavir in treatment naive-patients. PPIs are not recommended in treatment-experienced taking atazanavir.
- **dacomitinib**
pantoprazole will increase the level or effect of dacomitinib by unspecified interaction mechanism. Avoid or Use Alternate Drug. Concomitant use with a PPI decreases dacomitinib concentrations, which may reduce dacomitinib efficacy. Avoid use of PPIs with dacomitinib. As an alternative to PPIs, use locally-acting antacids or an H2-receptor antagonist. Administer at least 6 hours before or 10 hours after taking an H2-receptor antagonist.
- **darolutamide**
darolutamide will increase the level or effect of pantoprazole by Other (see comment). Avoid or Use Alternate Drug. Darolutamide is a BCRP inhibitor. Avoid coadministration with BCRP inhibitors. If use is unavoidable, closely monitor for adverse reactions and consider dose reduction of BCRP substrate drug (refer BCRP substrate prescribing information).

- **dasatinib**
pantoprazole will decrease the level or effect of dasatinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.
- **defactinib**
pantoprazole decreases levels of defactinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Defactinib exposure may be decreased when coadministered with proton pump inhibitors (PPIs). If concomitant use with PPIs cannot be avoided, separate administration by at least 2 hr.
- **digoxin**
pantoprazole will increase the level or effect of digoxin by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.
- **enasidenib**
enasidenib will increase the level or effect of pantoprazole by Other (see comment). Avoid or Use Alternate Drug. Avoid coadministration of enasidenib with sensitive OATP1B1, OATP1B3, or BCRP substrates, for which minimal concentration changes may lead to serious toxicities. If unavoidable, decrease the substrate(s) dosage in accordance with their Prescribing Information.
- **erlotinib**
pantoprazole decreases levels of erlotinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Coadministration of proton pump inhibitors and erlotinib may decrease erlotinib exposure and reduce its efficacy. Consider alternate acid reducing agents.
- **infigratinib (DSC)**
pantoprazole will decrease the level or effect of infigratinib (DSC) by inhibition of GI absorption. Applies only to oral form of both agents. Avoid or Use Alternate Drug.
- **itraconazole**
pantoprazole will decrease the level or effect of itraconazole by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.
- **ketoconazole**
pantoprazole will decrease the level or effect of ketoconazole by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.
- **leniolisib**
leniolisib will increase the level or effect of pantoprazole by Other (see comment). Avoid or Use Alternate Drug. Leniolisib, a BCRP inhibitor, may increase systemic exposure of BCRP substrates
- **levoketoconazole**
pantoprazole will decrease the level or effect of levoketoconazole by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.
- **lonafarnib**
pantoprazole will increase the level or effect of lonafarnib by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Avoid or Use Alternate Drug. If coadministration of lonafarnib (a sensitive CYP3A substrate) with weak CYP3A

inhibitors is unavoidable, reduce to, or continue lonafarnib at starting dose. Closely monitor for arrhythmias and events (eg, syncope, heart palpitations) since lonafarnib effect on QT interval is unknown.

lonafarnib will increase the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Avoid or Use Alternate Drug. Lonafarnib may increase the AUC and peak concentration of CYP2C19 substrates. If coadministration unavoidable, monitor for adverse reactions and reduce the CYP2C19 substrate dose in accordance with its approved product labeling.

- mesalamine
pantoprazole decreases effects of mesalamine by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Applies only to sustained release dosage form.
- neratinib
pantoprazole will decrease the level or effect of neratinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.
- nilotinib
pantoprazole decreases levels of nilotinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. As an alternative to PPIs, use H2-blockers ~10 hr before or ~2 hr after nilotinib, or use antacids ~2 hr before or ~2 hr after nilotinib (does not apply if using ODT nilotinib).
- nisoldipine
pantoprazole will increase the level or effect of nisoldipine by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.
- ozanimod
pantoprazole increases toxicity of ozanimod by Other (see comment). Avoid or Use Alternate Drug. Comment: Coadministration of ozanimod (a BCRP substrate) with BCRP inhibitors increases the exposure of the minor (RP101988, RP101075) and major active metabolites (CC112273, CC1084037) of ozanimod, which may increase the risk of ozanimod adverse reactions. .
- pazopanib
pantoprazole will decrease the level or effect of pazopanib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Avoid coadministration of pazopanib with drugs that raise gastric pH; consider short-acting antacids in place of PPIs and H2 antagonists; separate antacid and pazopanib dosing by several hours
- pexidartinib
pantoprazole will decrease the level or effect of pexidartinib by inhibition of GI absorption. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Avoid coadministration of proton pump inhibitors (PPIs) with pexidartinib. Use H2-receptor antagonists or antacids if needed. When using alternatives to PPIs, administer pexidartinib 2 hr before or after taking locally-acting antacids OR administer pexidartinib at least 2 hr before or 10 hr after taking an H2-receptor antagonist.
- rilzabrutinib

pantoprazole decreases levels of rilzabrutinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Coadministration may decrease rilzabrutinib exposure, resulting in reduced efficacy.

- **riociguat**
pantoprazole will increase the level or effect of riociguat by decreasing metabolism. Avoid or Use Alternate Drug. Coadministration of riociguat (substrate of CYP isoenzymes 1A1, 2C8, 3A, 2J2) with strong CYP inhibitors may require a decreased initial dose of 0.5 mg PO TID; monitor for signs of hypotension and reduce dose if needed
- **secretin**
pantoprazole, secretin. Other (see comment). Avoid or Use Alternate Drug.
Comment: Concomitant use of PPIs may cause a hyperresponse in gastrin secretion in response to stimulation testing with secretin, falsely suggesting gastrinoma. The time it takes for serum gastrin concentrations to return to baseline following discontinuation of PPIs is specific to the individual PPI. Following short-term treatment with pantoprazole, elevated gastrin levels return to normal by at least 3 months.
- **sofosbuvir/velpatasvir**
pantoprazole will decrease the level or effect of sofosbuvir/velpatasvir by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Velpatasvir solubility decreases as gastric pH increases (practically insoluble at pH >5). Coadministration of sofosbuvir/velpatasvir with omeprazole or other PPIs is not recommended. If considered medically necessary, give sofosbuvir/velpatasvir with food 4 hr before omeprazole 20 mg. Use with other PPIs has not been studied.
- **sotorasib**
pantoprazole will decrease the level or effect of sotorasib by inhibition of GI absorption. Applies only to oral form of both agents. Avoid or Use Alternate Drug. If use with an acid-reducing agent cannot be avoided, administer sotorasib 4 hr before or 10 hr after administration of a locally-acting antacid.
- **sparsentan**
pantoprazole decreases effects of sparsentan by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Proton pump inhibitors may decrease sparsentan exposure which may reduce efficacy of sparsentan.
- **taletrectinib**
pantoprazole decreases levels of taletrectinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Coadministration may decrease taletrectinib exposure and reduce its efficacy.
- **ziftomenib**
pantoprazole decreases levels of ziftomenib by unspecified interaction mechanism. Avoid or Use Alternate Drug. Proton pump inhibitors decrease zifomenib peak plasma concentrations by 70%.

Monitor Closely (78)

- **acalabrutinib**

acalabrutinib increases levels of pantoprazole by Other (see comment). Use Caution/Monitor. Comment: Acalabrutinib may increase exposure to coadministered BCRP substrates by inhibition of intestinal BCRP.

- amphetamine
pantoprazole, amphetamine. Other (see comment). Use Caution/Monitor. Comment: Reduced gastric acidity caused by proton pump inhibitors decreases time to Tmax for amphetamine and dextroamphetamine. AUC was unaffected. .
- ampicillin
pantoprazole will decrease the level or effect of ampicillin by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- apalutamide
apalutamide will decrease the level or effect of pantoprazole by increasing elimination. Use Caution/Monitor. Apalutamide weakly induces BCRP and may decrease systemic exposure of drugs that are BCRP substrates.
- belumosudil
pantoprazole will decrease the level or effect of belumosudil by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Increase belumosudil dosage to 200 mg PO BID when coadministered with proton pump inhibitors.
- berotralstat
pantoprazole increases levels of berotralstat by Other (see comment). Use Caution/Monitor. Comment: Monitor berotralstat (a BCRP substrate) for adverse effects when coadministered with a BCRP inhibitor.
- bortezomib
bortezomib will decrease the level or effect of pantoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.
- bosutinib
pantoprazole decreases levels of bosutinib by Other (see comment). Use Caution/Monitor. Comment: PPIs may decrease bosutinib concentration by ~45%; bosutinib displays pH-dependent solubility.
- budesonide
pantoprazole decreases effects of budesonide by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Enteric-coated budesonide dissolves at pH >5.5. Also, dissolution of extended-release budesonide tablets is pH dependent. Coadministration with drugs that increase gastric pH may cause these budesonide products to prematurely dissolve, and possibly affect release properties and absorption of the drug in the duodenum.
- cannabidiol
pantoprazole will increase the level or effect of cannabidiol by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Consider reducing the cannabidiol dose when coadministered with a moderate CYP2C19 inhibitor.

cannabidiol will increase the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Consider reducing

the dose of sensitive CYP2C19 substrates, as clinically appropriate, when coadministered with cannabidiol.

- **capecitabine**
pantoprazole decreases effects of capecitabine by unknown mechanism. Use Caution/Monitor. Retrospective data suggest elevated gastric pH caused by PPI use may impair capecitabine tablet dissolution and/or reduce absorption.
- **carbamazepine**
carbamazepine will decrease the level or effect of pantoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.
- **carbonyl iron**
pantoprazole will decrease the level or effect of carbonyl iron by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- **cenobamate**
cenobamate will increase the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Consider a dose reduction of CYP2C19 substrates, as clinically appropriate, when used concomitantly with cenobamate.
- **ceritinib**
pantoprazole will decrease the level or effect of ceritinib by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely.
- **cilostazol**
pantoprazole increases toxicity of cilostazol by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Consider decreasing cilostazol dose; moderate CYP2C19 inhibitors may increase serum levels of 3,4-dehydrocilostazol (active metabolite).
- **clopidogrel**
pantoprazole decreases effects of clopidogrel by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Clopidogrel efficacy may be reduced by drugs that inhibit CYP2C19. Inhibition of platelet aggregation by clopidogrel is entirely due to the active clopidogrel metabolite. Clopidogrel is metabolized in part by CYP2C19. Pantoprazole prescribing information state that coadministration with clopidogrel had no clinically important effect on exposure to clopidogrel active metabolite; no dose adjustment of clopidogrel is required .
- **crizotinib**
pantoprazole decreases levels of crizotinib by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor. Drugs that elevate the gastric pH may decrease the solubility of crizotinib and subsequently reduce its bioavailability. However, no formal studies have been conducted. .
- **cyclosporine**
pantoprazole, cyclosporine. Either increases toxicity of the other by Other (see comment). Use Caution/Monitor. Comment: When used for prolonged periods of time PPIs may cause hypomagnesemia and the risk is further increased when used concomitantly with drugs that also have the same effects.

- **dabrafenib**
pantoprazole will decrease the level or effect of dabrafenib by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor. Drugs that alter upper GI tract pH (eg, PPIs, H₂-blockers, antacids) may decrease dabrafenib solubility and reduce its bioavailability

dabrafenib will decrease the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Use alternative if available
- **danicopan**
danicopan will increase the level or effect of pantoprazole by Other (see comment). Use Caution/Monitor. Danicopan increases plasma concentrations of BCRP substrates; consider dose reduction of BCRP substrate according to its prescribing information.
- **dextroamphetamine**
pantoprazole, dextroamphetamine. Other (see comment). Use Caution/Monitor. Comment: Reduced gastric acidity caused by proton pump inhibitors decreases time to T_{max} for amphetamine and dextroamphetamine. AUC was unaffected. .
- **diazepam buccal**
pantoprazole will increase the level or effect of diazepam buccal by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Strong or moderate CYP2C19 inhibitors may decrease rate of diazepam elimination, thereby increasing adverse reactions to diazepam.
- **diazepam intranasal**
pantoprazole will increase the level or effect of diazepam intranasal by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Strong or moderate CYP2C19 inhibitors may decrease rate of diazepam elimination, thereby increasing adverse reactions to diazepam.
- **digoxin**
pantoprazole increases toxicity of digoxin by Other (see comment). Use Caution/Monitor. Comment: Prolonged use of PPIs may cause hypomagnesemia and increase risk for digoxin toxicity.
- **dordaviprone**
dordaviprone and pantoprazole both increase QTc interval. Use Caution/Monitor. Increase the frequency of ECG and electrolyte monitoring.
- **duvelisib**
pantoprazole will increase the level or effect of duvelisib by Other (see comment). Use Caution/Monitor. Coadministration of duvelisib (a BCRP substrate) with a BCRP transport inhibitor may increase levels or effects of duvelisib.
- **elagolix**
elagolix will increase the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Elagolix is a weak CYP2C19 inhibitor. Caution with sensitive CYP2C19 substrates.
- **encorafenib**

encorafenib will increase the level or effect of pantoprazole by Other (see comment). Modify Therapy/Monitor Closely. Encorafenib (a BCRP inhibitor) may increase the concentration and toxicities of BCRP substrates. Closely monitor for signs and symptoms of increased exposure and consider adjusting the dose of these substrates.

- eslicarbazepine acetate
eslicarbazepine acetate will increase the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.
- ferric gluconate
pantoprazole will decrease the level or effect of ferric gluconate by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- ferric maltol
pantoprazole will decrease the level or effect of ferric maltol by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- ferrous fumarate
pantoprazole will decrease the level or effect of ferrous fumarate by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- ferrous gluconate
pantoprazole will decrease the level or effect of ferrous gluconate by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- ferrous sulfate
pantoprazole will decrease the level or effect of ferrous sulfate by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- fexinidazole
fexinidazole will increase the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.
- finerenone
pantoprazole will increase the level or effect of finerenone by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Modify Therapy/Monitor Closely. Monitor serum potassium during initiation and dosage adjustment of either finerenone or weak CYP3A4 inhibitors. Adjust finerenone dosage as needed.
- fluvoxamine
fluvoxamine will increase the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.
- fosamprenavir
pantoprazole will decrease the level or effect of fosamprenavir by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- fostemsavir
fostemsavir will increase the level or effect of pantoprazole by Other (see comment). Modify Therapy/Monitor Closely. Fostemsavir inhibits BCRP

transporters. If possible, avoid coadministration or modify dose of BCRP substrate coadministered with fostemsavir.

- **gefitinib**
pantoprazole decreases levels of gefitinib by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor. Avoid coadministration of gefitinib with PPIs if possible. If treatment with a PPI is required, separate gefitinib and PPI doses by 12 hr.
- **glecaprevir/pibrentasvir**
pantoprazole will increase the level or effect of glecaprevir/pibrentasvir by decreasing metabolism. Use Caution/Monitor. Caution when coadministering glecaprevir/pibrentasvir with P-gp/BCRP inhibitors.

glecaprevir/pibrentasvir will increase the level or effect of pantoprazole by Other (see comment). Use Caution/Monitor. Glecaprevir/pibrentasvir may increase plasma concentration of P-gp and BCRP substrates.
- **iron dextran complex**
pantoprazole will decrease the level or effect of iron dextran complex by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- **iron sucrose**
pantoprazole will decrease the level or effect of iron sucrose by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- **isavuconazonium sulfate**
pantoprazole will increase the level or effect of isavuconazonium sulfate by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.
- **itraconazole**
itraconazole will increase the level or effect of pantoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Modify Therapy/Monitor Closely. For patients using the Sporanox brand of itraconazole (capsules or solution), administer proton pump inhibitors at least 2 hr before or 2 hr after itraconazole. Use of Sporanox oral solution or administration of itraconazole with an acidic beverage (eg, cola) may minimize the significance of this interaction.
- **ledipasvir/sofosbuvir**
pantoprazole decreases levels of ledipasvir/sofosbuvir by Other (see comment). Use Caution/Monitor. Comment: Ledipasvir solubility decreases as pH increases; drugs that increase gastric pH are expected to decrease levels of ledipasvir; proton-pump inhibitor doses comparable to omeprazole <20 mg can be administered simultaneously with ledipasvir/sofosbuvir under fasted conditions.
- **letermovir**
letermovir increases levels of pantoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor. Monitor and dose adjustment may be necessary.
- **lumacaftor/ivacaftor**
lumacaftor/ivacaftor, pantoprazole. affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. In vitro studies suggest that lumacaftor may induce and

ivacaftor may inhibit CYP2C19 substrates. .

lumacaftor/ivacaftor will decrease the level or effect of pantoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

Lumacaftor/ivacaftor is a strong CYP3A4 inhibitor and also has the potential to induce CYP2C19 and both induce and inhibitor P-gp.

- methotrexate
pantoprazole increases levels of methotrexate by decreasing renal clearance. Use Caution/Monitor. Increased risk of toxicity with higher doses.
- methylphenidate
pantoprazole decreases effects of methylphenidate by enhancing GI absorption. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Since the characteristics of methylphenidate extended release capsules (Ritalin LA) are pH dependent, coadministration of antacids or acid suppressants could alter the release of methylphenidate. Consider separating the administration of the antacid and the methylphenidate extended-release capsules may be avoided.
- mitapivat
mitapivat decreases levels of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Monitor patients for loss of therapeutic effect if taking sensitive CYP2C19 substrates. .
- momelotinib
momelotinib increases toxicity of pantoprazole by plasma protein binding competition. Modify Therapy/Monitor Closely. Momelotinib (BCRP inhibitor) may increase exposure of BCRP substrates, which may increase the risk of BCRP substrate adverse reactions. Dose adjustment of other BCRP substrates may necessary.
- mycophenolate
pantoprazole will decrease the level or effect of mycophenolate by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Potential interaction applies to the prodrug mycophenolate mofetil conversion to active mycophenolic acid. Enteric coated mycophenolate sodium formulation is less sensitive to this interaction.
- oteseconazole
oteseconazole will increase the level or effect of pantoprazole by Other (see comment). Modify Therapy/Monitor Closely. Otesezonale, a BCRP inhibitor, may increase the effects and risk of toxicities of BCRP substrates. Use lowest starting dose of BCRP substrate, or consider reducing BCRP substrate dose.
- paltusotine
pantoprazole decreases levels of paltusotine by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Coadministration demonstrated a dose-dependent decrease in paltusotine exposure owing to decreased paltusotine solubility at elevated gastric pH. May require increased paltusotine dose. Avoid coadministration if patient is already taking paltusotine 60 mg.
- phenytoin

phenytoin will decrease the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- polysaccharide iron
pantoprazole will decrease the level or effect of polysaccharide iron by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- posaconazole
pantoprazole will decrease the level or effect of posaconazole by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- regorafenib
regorafenib will increase the level or effect of pantoprazole by Other (see comment). Modify Therapy/Monitor Closely. Regorafenib likely inhibits BCRP (ABCG2) transport. Coadministration with a BCRP substrate may increase systemic exposure to the substrate and related toxicity.
- revumenib
revumenib and pantoprazole both increase QTc interval. Use Caution/Monitor. ECG monitoring recommended if concomitant use can't be avoided and withhold therapy if QTc interval is >480 msec; restart therapy after QTc interval returns to <480 msec
- rifampin
rifampin will decrease the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.
- rose hips
pantoprazole will decrease the level or effect of rose hips by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- rucaparib
rucaparib will increase the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Adjust dosage of CYP2C19 substrates, if clinically indicated.
- safinamide
safinamide will increase the level or effect of pantoprazole by Other (see comment). Use Caution/Monitor. Safinamide and its major metabolite may inhibit intestinal BCRP. Monitor BCRP substrates for increased pharmacologic or adverse effects.
- selexipag
pantoprazole will increase the level or effect of selexipag by Other (see comment). Use Caution/Monitor. Selexipag is a ABCG2 (BCRP) substrate. Monitor selexipag for increased pharmacologic or adverse effects when coadministered with ABCG2 (BCRP) inhibitors.
- sparsentan
sparsentan will decrease the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Sparsentan (a CYP2C19 inducer) decreases exposure of CYP2C19 substrates and reduces efficacy related to these substrates.

- St John's Wort
St John's Wort will decrease the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.
- stiripentol
stiripentol will increase the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Consider reducing the dose of CYP2C19 substrates, if adverse reactions are experienced when administered concomitantly with stiripentol.

stiripentol will increase the level or effect of pantoprazole by Other (see comment). Modify Therapy/Monitor Closely. Stiripentol is a BCRP transport inhibitor. Consider dosage reduction for BCRP substrates if adverse effects are experienced when coadministered.
- tacrolimus
pantoprazole will increase the level or effect of tacrolimus by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Concomitant administration may increase tacrolimus whole blood concentrations, particularly in intermediate or poor metabolizers of CYP2C19
- tafamidis
tafamidis will increase the level or effect of pantoprazole by Other (see comment). Use Caution/Monitor. Tafamidis inhibits breast cancer resistant protein (BCRP) in vitro and may increase exposure of BCRP substrates following tafamidis or tafamidis meglumine administration. Dosage adjustment of these BCRP substrates may be necessary.
- taletrectinib
taletrectinib and pantoprazole both increase QTc interval. Use Caution/Monitor. Concomitant use may increase the risk of QTc interval prolongation. If concomitant use is unavoidable, adjust the frequency of ECG and electrolyte monitoring.
- tecovirimat
tecovirimat will increase the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Tecovirimat is a weak inhibitor of CYP2C8 and CYP2C19. Monitor for adverse effects if coadministered with sensitive substrates of these enzymes.
- theophylline
pantoprazole increases toxicity of theophylline by Other (see comment). Use Caution/Monitor. Comment: Prolonged use of proton pump inhibitors can cause hypochlorhydria, which in turn causes peristalsis in small intestine to increase and peristalsis in the proximal colon to decrease; monitor for toxicity.
- triclabendazole
triclabendazole will increase the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.
- vadadustat
vadadustat will increase the level or effect of pantoprazole by Other (see comment). Use Caution/Monitor. Vadadustat may increase exposure of BCRP

substrates. Monitor for signs of adverse effect of BCRP substrate and reduce substrate dose in accordance with their product labeling.

- voriconazole
voriconazole will increase the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

voriconazole will increase the level or effect of pantoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- ziftomenib
pantoprazole, ziftomenib. QTc interval. Use Caution/Monitor. Caution if ziftomenib is coadministered with drugs that exacerbated risk of QT prolongation (eg, hypokalemia, orthostatic hypotension, inhibits metabolism of ziftomenib).

Minor (28)

- blessed thistle
blessed thistle decreases effects of pantoprazole by pharmacodynamic antagonism. Minor/Significance Unknown. Theoretical interaction.
- clobazam
pantoprazole will increase the level or effect of clobazam by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown. Dosage adjustment may be required; CYP2C19 inhibitors may result in increased exposure to N-desmethylclobazam (active metabolite).
- cyanocobalamin
pantoprazole decreases levels of cyanocobalamin by inhibition of GI absorption. Applies only to oral form of both agents. Minor/Significance Unknown.
- darifenacin
darifenacin decreases effects of pantoprazole by Other (see comment). Minor/Significance Unknown. Comment: Effectiveness of proton pump inhibitors may be decreased theoretically if administered with other antisecretory agents .
- devil's claw
devil's claw decreases effects of pantoprazole by pharmacodynamic antagonism. Minor/Significance Unknown.
- dexamethasone
dexamethasone will decrease the level or effect of pantoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
- dronedarone
dronedarone will increase the level or effect of pantoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
- efavirenz
efavirenz will increase the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.

- elvitegravir/cobicistat/emtricitabine/tenofovir DF
elvitegravir/cobicistat/emtricitabine/tenofovir DF, pantoprazole. affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
Based on drug interaction studies conducted with the components of Stribild, no clinically significant drug interactions have been either observed or are expected when coadministered with PPIs.
- eslicarbazepine acetate
eslicarbazepine acetate will decrease the level or effect of pantoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
Monitor for GI symptoms; net increased or decreased effect on PPI action unclear due to opposing CYP450 actions.
- etravirine
etravirine will decrease the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.
- ferric carboxymaltose
pantoprazole will decrease the level or effect of ferric carboxymaltose by increasing gastric pH. Applies only to oral form of both agents. Minor/Significance Unknown.
- fluconazole
fluconazole will increase the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.
- fosphenytoin
fosphenytoin will decrease the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.
- ketoconazole
ketoconazole will increase the level or effect of pantoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

ketoconazole will increase the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.
- levoketoconazole
levoketoconazole will increase the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.

levoketoconazole will increase the level or effect of pantoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
- levothyroxine
pantoprazole decreases levels of levothyroxine by increasing gastric pH. Applies only to oral form of both agents. Minor/Significance Unknown. Conflicting evidence regarding this interaction exists.
- liothyronine
pantoprazole decreases levels of liothyronine by increasing gastric pH. Applies only to oral form of both agents. Minor/Significance Unknown. Conflicting evidence regarding this interaction exists.

- **liotrix**
pantoprazole decreases levels of liotrix by increasing gastric pH. Applies only to oral form of both agents. Minor/Significance Unknown. Conflicting evidence regarding this interaction exists.
- **lisdexamfetamine**
pantoprazole, lisdexamfetamine. Other (see comment). Minor/Significance Unknown. Comment: Reduced gastric acidity caused by proton pump inhibitors decreases time to Tmax for amphetamine and dextroamphetamine. AUC was unaffected. .
- **methamphetamine**
pantoprazole decreases levels of methamphetamine by Other (see comment). Minor/Significance Unknown. Comment: Time to maximum concentration (Tmax) of amphetamine is decreased compared to when administered alone; monitor patients for changes in clinical effect and adjust therapy based on clinical response.
- **modafinil**
modafinil will increase the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown. Net effect on pantoprazole actions unknown due to opposing effects of CYP450 enzymes; monitor
- **oxcarbazepine**
oxcarbazepine will increase the level or effect of pantoprazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown. Unclear on net effect of pantoprazole action due to opposing effects by CYP450 enzymes; monitor
- **pentobarbital**
pentobarbital will decrease the level or effect of pantoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
- **phenobarbital**
phenobarbital will decrease the level or effect of pantoprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.
- **phytoestrogens**
pantoprazole decreases levels of phytoestrogens by inhibition of GI absorption. Applies only to oral form of both agents. Minor/Significance Unknown.
- **ponatinib**
pantoprazole decreases levels of ponatinib by increasing gastric pH. Applies only to oral form of both agents. Minor/Significance Unknown.
- **thyroid desiccated**
pantoprazole decreases levels of thyroid desiccated by increasing gastric pH. Applies only to oral form of both agents. Minor/Significance Unknown. Conflicting evidence regarding this interaction exists.

Adverse Effects

1-10%

Headache (>4%)

Abdominal pain (4%)

Facial edema (<4%)

Generalized edema (<2%)

Chest pain (4%)

Diarrhea (4%)

Constipation (<4%)

Pruritus (4%)

Rash (4%)

Flatulence (<4%)

Hyperglycemia (1%)

Nausea (1%)

Vomiting (>4%)

Photosensitivity (<2%)

Frequency Not Defined

Angioedema

Atrophic gastritis

Anterior ischemic optic neuropathy

Hepatocellular damage leading to hepatic failure

Interstitial nephritis

Pancreatitis

Pancytopenia

Rhabdomyolysis

Risk of anaphylaxis

Stevens-Johnson syndrome

Fatal toxic epidermal necrolysis

Erythema multiforme

Postmarketing Reports

Asthenia, fatigue, malaise

Hepatocellular damage leading to jaundice and hepatic failure

Anaphylaxis (including anaphylactic shock)

Agranulocytosis, pancytopenia

Taste disorders (ageusia, dysgeusia)

Investigations: Weight changes

Hyponatremia, hypomagnesemia

Rhabdomyolysis, bone fracture

Hallucination, confusion, insomnia, somnolence

Interstitial nephritis

Severe dermatologic reactions (some fatal), including erythema multiforme, Stevens-Johnson syndrome, and toxic epidermal necrolysis (some fatal), as well as angioedema (Quincke edema), drug reaction with eosinophilia and systemic symptoms (DRESS), acute generalized exanthematous pustulosis (AGEP)

Cutaneous and systemic lupus erythematosus

Cyanocobalamin (vitamin B-12) deficiency

Clostridium difficile-associated diarrhea

Fundic gland polyps

Acute tubulointerstitial nephritis

Hypocalcemia

Erectile dysfunction

Warnings

Contraindications

Hypersensitivity to pantoprazole or other proton pump inhibitors (PPIs)

Concomitant administration with rilpivirine containing products

Cautions

PPIs are possibly associated with increased incidence of Clostridium difficile-associated diarrhea (CDAD); consider diagnosis of CDAD for patients taking PPIs who have diarrhea that does not improve

Severe hepatic impairment

Cutaneous lupus erythematosus (CLE) and systemic lupus erythematosus (SLE) reported with PPIs; avoid using for longer than medically indicated; discontinue if signs or

symptoms consistent with CLE or SLE are observed and refer patient to specialist; most patients improve with discontinuation of PPI alone in 4-12 weeks; serological testing (e.g. ANA) may be positive and elevated serological test results may take longer to resolve than clinical manifestations

Severe cutaneous adverse reactions, including Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS), and acute generalized exanthematous pustulosis (AGEP) reported in association with use of PPIs; discontinue therapy at first signs or symptoms of severe cutaneous adverse reactions or other signs of hypersensitivity and consider further evaluation

Published observational studies suggest that PPI therapy may be associated with an increased risk for osteoporosis-related fractures of the hip, wrist, or spine, particularly with prolonged (>1 year), high-dose therapy

Decreased gastric acidity increases serum chromogranin A (CgA) levels and may cause false-positive diagnostic results for neuroendocrine tumors; temporarily discontinue PPIs before assessing CgA levels

PPIs may decrease the efficacy of clopidogrel by reducing the formation of the active metabolite

Therapy increases risk of Salmonella, Campylobacter, and other infections

Infusion related reactions including thrombophlebitis and hypersensitivity reported

Daily long-term use (e.g., longer than 3 years) may lead to malabsorption or a deficiency of cyanocobalamin

Acute interstitial nephritis reported in patients taking proton pump inhibitors

Relief of symptoms does not eliminate the possibility of a gastric malignancy; consider additional follow-up and diagnostic testing in adult patients who have suboptimal response or early symptomatic relapse after completing treatment with a PPI

Risk of salmonella and campylobacter infections increased with use of proton pump inhibitors

May elevate and/or prolong serum concentrations of methotrexate and/or its metabolite when administered concomitantly with PPIs, possibly leading to toxicity; consider a temporary withdrawal of PPI therapy with high dose methotrexate administration

PPI therapy is associated with increased risk of fundic gland polyp; risk increases with long-term use >1 year; patient may be asymptomatic; problem usually identified incidentally on endoscopy; use shortest duration of therapy appropriate to condition being treated

Acute tubulointerstitial nephritis (TIN) reported in patients taking PPIs; may occur at any point during PPI therapy; patients may present with varying signs and symptoms from symptomatic hypersensitivity reactions to non-specific symptoms of decreased renal function (eg, malaise, nausea, anorexia); in reported case series, some patients were diagnosed on biopsy and in absence of extra-renal manifestations (eg, fever, rash or arthralgia); discontinue therapy and evaluate patients with suspected acute TIN

Hypomagnesemia and mineral metabolism

- Hypomagnesemia may occur with prolonged use (>1 year); adverse effects may result, including tetany, arrhythmias, and seizures; in 25% of cases reviewed, magnesium supplementation alone did not improve low serum magnesium levels, and the PPI had to be discontinued; consider monitoring magnesium levels prior to initiation of PPI treatment and periodically
- Hypomagnesemia may lead to hypocalcemia and/or hypokalemia and may exacerbate underlying hypocalcemia in at-risk patients; in most patients, treatment of hypomagnesemia required magnesium replacement and discontinuation of the PPI
- Consider monitoring magnesium and calcium levels prior to initiation of therapy and periodically while on treatment in patients with a preexisting risk of hypocalcemia (eg, hypoparathyroidism); supplement with magnesium and/or calcium as necessary. If hypocalcemia is refractory to treatment, consider discontinuing the PPI

Pregnancy & Lactation

Pregnancy

There are no adequate and well-controlled studies in pregnant women; advise pregnant women of the potential risk of fetal harm; drug should be used during pregnancy only if potential benefit justifies potential risk to fetus

Lactation

Pantoprazole and metabolites are excreted in milk of rats. Pantoprazole excretion in human milk has been detected in a study of a single nursing mother after a single 40 mg oral dose of pantoprazole sodium; clinical relevance of this finding not known; many drugs excreted in human milk have potential for serious adverse reactions in nursing infants; based on potential for tumorigenicity shown for pantoprazole sodium in rodent carcinogenicity studies, a decision should be made whether to discontinue nursing or to discontinue drug, taking into account benefit of drug to mother

Pregnancy Categories

A: Generally acceptable. Controlled studies in pregnant women show no evidence of fetal risk.

B: May be acceptable. Either animal studies show no risk but human studies not available or animal studies showed minor risks and human studies done and showed no risk.

C: Use with caution if benefits outweigh risks. Animal studies show risk and human studies not available or neither animal nor human studies done.

D: Use in LIFE-THREATENING emergencies when no safer drug available. Positive evidence of human fetal risk.

X: Do not use in pregnancy. Risks involved outweigh potential benefits. Safer alternatives exist.

NA: Information not available.

Pharmacology

Mechanism of Action

PPI; binds to H⁺/K⁺-exchanging ATPase (proton pump) in gastric parietal cells, resulting in blockage of acid secretion

Absorption

Bioavailability: 77% (PO; neither food nor antacid alters bioavailability)

Onset (PO PUD): 24 hr (initial response)

Duration (PO at steady state): 7 days (PUD)

Peak plasma time: 2.8 hr (PO); at end of infusion (IV)

Distribution

Protein bound: 98%

Vd: 11-24 L

Metabolism

Metabolized extensively by hepatic P450 enzyme CYP2C19; second pathway through CYP3A4

Slow metabolizers (3% of Caucasians and African Americans) are deficient in CYP2C19 enzyme system; plasma concentration can increase by 5-fold or more in comparison with that found in persons who have the enzyme

Metabolites: Desmethylpantoprazole sulfate conjugate (activity unknown)

Elimination

Half-life: 1 hr; increased to 3.5-10 hr with CYP2C19 deficiency

Dialyzable: No

Renal clearance: 0.1 L/hr/kg

Total body clearance: 7.6-14 L/hr

Excretion: Urine (71%); feces (18%)

Administration

Oral Administration

Switch patients from IV to PO as soon as able to swallow

Tablets

- May take with or without food

- Concomitant administration of antacids does not affect tablet absorption
- Swallow tablet whole; do not split, crush, or chew

Oral suspension

- Administer ~30 minutes before meals
- Do not split, crush, or chew
- Oral suspension packet is a fixed dose and cannot be divided to make a smaller dose
- Administer suspension in apple juice or applesauce; do not administer in water, other liquids, or foods
- NG tube
 - For patients with a nasogastric (NG) or gastrostomy tube, oral suspension can be administered with apple juice

Missed dose

- Take a missed dose as soon as possible
- If it is almost time for the next dose, skip missed dose and take the next dose at the regular scheduled time
- Do not take 2 doses at the same time

IV Incompatibilities

Y-site: Midazolam, zinc solutions

IV Compatibilities

Y-site: D5W, 0.9% NaCl, or LR

IV Preparation

Injectable solution

- Thaw frozen container at room temperature 20-25°C (68-77°F) or under refrigeration at 2-8°C (36-46°F)
- Product should not be thawed by immersion in water baths or by microwave irradiation
- Do not force thaw
- No further dilution necessary
- Check for minute leaks by squeezing container firmly; if leaks detected, discard solution as sterility may be impaired
- Do not add supplementary medication
- Inspect thawed solution visually for particulate matter before and during administration
- Thawed solution is expected to range from colorless to yellow over time

Powder for reconstitution (GERD)

- Reconstitute with 10 mL 0.9% NaCl
- Dilute reconstituted solution to final concentration
 - Pediatric patients aged 3 months to <1 year: Dilute with 21 mL 0.9% NaCl to final concentration of ~1.3 mg/mL
 - Adults and pediatric patients aged 1-17 years: Dilute with 100 mL D5W or 0.9% NaCl to final concentration of ~0.4 mg/mL

Powder for reconstitution (ZE)

- 15-min infusion: Reconstitute each vial with 10 mL 0.9% NaCl, THEN
- Combine 2 vials and further dilute with 80 mL D5W, 0.9% NaCl, or LR to total volume of 100 mL (concentration 0.8 mg/mL)
- 2-min injection: Reconstitute with 10 mL 0.9% NaCl to final concentration of 4 mg/mL

IV Administration

Infuse IV over ~15 minutes

Administer IV through a dedicated line or through a Y-site

Flush IV line before and after administration with either D5W or 0.9% NaCl

When administered through a Y-site, immediately stop use if precipitation or discoloration occurs

Storage**Tablets or granules for oral suspension**

- Store at 20-25°C (68-77°F); excursions permitted to 15-30°C (59-86°F)

Injection, solution

- Freeze at or below -20°C (-4°F)
- Protect from light
- Thawed solution
 - Refrigerate at 2-8°C (36-46°F) for up to 21 days, OR
 - Room temperature at 20-25°C (68-77°F) for up to 48 hr
 - Protect from light
 - Do not refreeze once thawed

Injection, powder for reconstitution

- Refrigerate unopened vials at 2-8°C (36-46°F); protect from light
- Reconstituted solution: Store for up to 6 hr at room temperature before further dilution

- Diluted solution: Store for up to 24 hr at room temperature before administration
- Do not freeze reconstituted or diluted solution

Formulary

Adding plans allows you to compare formulary status to other drugs in the same class.

Formulary information is available for health plans only in the United States.

Medscape prescription drug monographs are based on FDA-approved labeling information, unless otherwise noted, combined with additional data derived from primary medical literature.
